

A Critical Evaluation on Lovelace's Objection to Artificial Intelligence

In 2001, the Lovelace test, named in the honor of Lady Lovelace, was created as a method to evaluate modern computing systems by testing whether a system can produce an output that humans cannot account for (*Can Computers Surprise Us?* | *Ada Lovelace Institute*, n.d.). Lady Lovelace argued in her foundational writings on the Analytical Engine that a machine can only be credited with genuine intelligence if it originates something independently. Artificial intelligence (AI) describes the field of technology that seeks to simulate human intelligence in areas including learning, problem solving, decision making, and even creativity. At a time where AI is widely regarded as rapidly pursuing AGI¹ and competing with the capabilities of human cognition, in this paper, I will posit that although modern generative AI produces outputs that can appear as creative and novel, Lovelace's argument still stands—genuine origination requires a conscious self capable of subjective experience and inherent intent.

Before evaluating Lovelace's claim, I will first clearly define and distinguish two terms central to the argument: creativity and origination. Creativity is defined as a property of the novel output that is produced, and is independent of where the output came from or what drove its production. Origination, the term explicitly used in Lovelace's claim, refers to the origin or source of an idea or work that creates a new, unprecedented result. Origination asks whether the agent produced an output from its own resources rather than executing prior instruction. Although creativity can suggest originality, these concepts are different because something can be creative without being originated. For example, a computer's novel output could be attributed and traced back to the programmer instead of the machine itself. Therefore, this distinction is

¹ Artificial General Intelligence (AGI) is proposed as a theoretical advancement of AI with superhuman level ability to apply knowledge across a wide range of tasks and domains.

important to the paper's argument as the question is not necessarily whether AI can produce creative outputs, but whether it can originate them.

Ada Lovelace was born Augusta Ada Byron, and she worked under Charles Babbage as an assistant, translating and adding her own extensive notes to Luigi Menabrea's paper, to publish "Sketch of the Analytical Engine, Invented by Charles Babbage". In 1843, Ada Lovelace published her argument on the limitations of computing machines where she wrote that the "Analytical Engine has no pretensions whatever to originate anything" and is limited to "whatever we know how to order it to perform" (Lovelace, 1843). The Analytical Engine, proposed by Charles Babbage in 1837, was a design for a general-purpose, programmable, steam-powered computing machine that incorporated arithmetic logic, conditional branching, loops and integrated memory. Lovelace thought the Analytical Engine was revolutionary, believing the engine could represent numbers as symbols or other abstract items, and thus theoretically compute anything the programmer wanted. The Stanford Encyclopedia of Philosophy notes how Lovelace "foresaw the possibility of using the Analytical Engine for non-numeric computation," and she anticipated that such a machine might one day compose music (*The Modern History of Computing (Stanford Encyclopedia of Philosophy)*, n.d.). Yet, even this technology would require a human programmer to specify every step, aligning to her central claim.

One significant challenge to Lovelace's position comes from Turing, who argued that the individual self underlying human creativity is not different from a machine's prior inputs. In the mid-1900s, Alan Turing famously disagreed with Lovelace's objection, arguing that there was no human faculty that could not one day be replicated by computers. He questioned the inability of computers to produce original work, asking: "Who can be certain that 'original work' that he has

done was not simply the growth of the seed planted in him by teaching, or the effect of following well-known general principles” (Turing, 1950). He challenged Lovelace’s position by suggesting that human beings are also shaped entirely by prior inputs of life experiences, meaning that the distinction between human and machine origination is unclear. However, this argument fails because it conflates these inputs with the concept of authoring an output. Human ideas are shaped by prior experience, and this requires a self that synthesises these experiences—a person consciously or unconsciously chooses which experiences to draw on, processes them through their personal values, and directs output towards a purpose that is their own. Advanced computing machines, such as current Large Language Models (LLMs), have no such self as they do not have inherent, personal intentions. The inputs simply pass through it, and even if human creativity is based on intrinsic inputs, true originality comes from the product of a conscious subject. Turing’s argument fails to undermine the distinction between executing a process and originality being a product of intrinsic motivation and sense of self.

It may be easy to conclude that the current capabilities of modern generative AI disproves Lovelace’s objection that machines cannot originate anything. For example, researchers at the University of New South Wales used a ChatGPT-like chatbot to create a patent idea. It designed a disposable glove with a fractal grip, and a database search confirmed the idea is unique and patentable. However, the AI required a human prompt to produce the output, and every parameter of its generative behavior was defined by its designers and training data. This aligns with Lovelace’s claim that the technology has no inherent intent or desire to act upon anything. In another project, deep-learning researcher Louis Bradshaw programmed an AI model called Aria to compose and evaluate its own piano pieces (Bradshaw et al., n.d.). Aria can vary musical parameters, such as the pitch of a note, and provides comments on why its choices—in line with

the novel musical production *Lovelace* anticipated. Despite this, Aria's capacity is specified by its architecture and training, which means that its outputs are the products of the programmer's origination.

Empirical research into the quality and character of AI-generated creative work supports that without an internal self to draw on, AI consistently falls short wherever creativity comes from personal expression. From poetry and video to scientific ideas and music, AI-generated content now rivals many human-made works, and the definition of creativity is getting blurred in this landscape. Although generative AI can create novel things, it seems that humans still have the edge over machines. A study from Montreal compared short stories written by humans versus pieces generated by LLMs. While some AI-generated stories were judged to be as good as attempts by amateur human writers, experts rated the AI stories as much poorer in quality than professional stories published in *The New Yorker*. Critics claimed that the AI-generated stories lacked “narrative endings, rhetorical complexity and character development,” and these storytelling qualities largely come from synthesizing lived experiences and rely on individual expression to tell meaningful stories in these aspects (Doshi & Hauser, 2024). Another study describes an experiment comparing humans and AI models with dreaming up new functions for everyday objects, LLMs couldn't match the innovative capacity of a group of five-year-old children (Stevenson et al., 2022). These results corroborate with the earlier claims that because computing machines are bound to perform operations explicitly programmed by humans and do not have an inherent sense of self associated with intention and imagination, which limits their ability to originate and output truly novel results.

Evidence from AI-driven scientific exploration reinforces that the creative process requires entities capable of novel insights, and this skill is limited in AI possession—for now.

Researchers from the Emylon Business School in Lyon, France, describe how AI models struggle with the imaginative leaps required to generate truly new insights in science. In their study, chatbots proposed fewer hypotheses and conducted fewer experiments and failed to revise hypotheses or conduct any new experiments based on their findings. From this, it was concluded that ChatGPT-4 does not have the necessary creative spark to notice and interpret anomalous results or to ask surprising and important questions. Although these results are limited by the existing capabilities of the AI technologies, is it just a matter of technical depth that enables this creativity? James Kaufman, an educational psychologist at the University of Connecticut, argues that we need to understand the process of creating rather than just looking at the end result, claiming that “AI can produce a creative product, sure, [b]ut it doesn’t go through a creative process,” and thus, AI is not truly a creative entity (Runco, 2023). Therefore, it can be argued that it is the process, not the result, that constitutes genuine origination in Lovelace’s argument.

Lovelace’s objection to the possibilities of artificial intelligence seem to align more strongly with today’s computing landscape than initially anticipated. Modern AI demonstrates that machines can produce outputs that are novel and in many contexts, difficult to distinguish from human work, supporting Turing’s predictions on the unfixed capabilities of machines. Yet, this does not refute Lovelace’s deeper point. The creativity that AI appears to exhibit is the product of prompted input data and ignores the vital, intangible qualities that humans use in their creative output. These neural networks follow algorithms, while people use subjective emotions, values, and lived experiences to imagine and make creative decisions that are not reducible to prior instruction. The modern debate echoes the issue Turing raised, but evidence suggests Lovelace was correct in her assertion about origination defined as the capacity to create rather than simply execute. Until a machine can be said to have a conscious self from which it acts

upon intrinsic intent, desires, and *lived* experiences, origination still remains a uniquely human faculty, and the appearance of AI creativity does not change this underlying fact.

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